LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Supplementary protective material and their uses.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Supplementary protective material and their uses.
5. Specific Objectives	6.1.2.1 Trainee will be able to explain various corrosion prevention measures.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of application of supplementary protective materials on test panels; Safety: Wear gloves and eye protection; Supervision: Instructor monitors material handling and disposal.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary cause of corrosion in aircraft structures?
(b) Name one common method used to prevent corrosion on aircraft surfaces.
(c) Why is it important to apply supplementary protective materials before painting?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of supplementary protective materials • Role of primers in corrosion prevention • Types of sealants and their applications • Function of anti-corrosive paints • Importance of surface preparation	L & Ds	PPT, IB	15 min
(b)	• Application techniques for sealants • Inspection criteria for protective coatings • Maintenance schedules for supplementary materials • Removal and replacement procedures • Storage and handling of protective materials	L & Ds	PPT, PDF	20 min
(c)	• Common defects in protective coatings • Troubleshooting application errors • Impact of environmental factors on materials • Documentation of maintenance activities • Safety precautions during application	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a primer and a topcoat in corrosion prevention.
(b) Describe the steps involved in inspecting a sealant application for defects.
(c) How do environmental conditions affect the longevity of supplementary protective materials?

12. Home Assignment and Follow Up	Prepare a brief report on the periodicity of cleaning and its relation to the effectiveness of supplementary protective materials, referencing the upcoming lesson on aircraft cleaning.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aircraft cleaning, periodicity of cleaning, various methods used.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min